Lawrence Smith

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Research Interests

Computational Design for Climate Technology Existential challenges require our sharpest tools. I drive the automatic production of solutions to challenging mechanical design problems, specifically in nonlinear design spaces where human intuition struggles. Core contributions include compact and flexible design representations, accelerated numerical models of physical phenomena, and synthesis of design, simulation, and fabrication processes.

Education

Ph.D. & M.S. Mechanical Engineering University of Colorado Boulder — 2023 & 2019

B.S. Mechanical Engineering California State Polytechnic University, San Luis Obispo — 2014

Experience

Postdoctoral Researcher Max Planck Institute for Intelligent Systems — 2024–2027 (est.)
Robotic Materials Department; Supervised by Christoph Keplinger

- Awarded the Alexander von Humboldt Postdoctoral Fellowship for research into harvesting ocean wave energy using electrostatics
- Publication of electrostatic energy generator units with field-leading performance, *Advanced Science*
- Led simulation efforts in investigation of vibrotactile sensing in functionally graded whiskers, *Science*

Doctoral Candidate & Researcher University of Colorado, Boulder — 2018–2023
Supervised by Robert MacCurdy, dissertation on computational design and fabrication of soft structures

- Award winning conference proceeding and subsequent IEEE journal publication presenting open-source toolkit for seamless design, simulation, and fabrication of pneumatic soft actuators
- Designed tunable energy-absorbing metamaterials which transmit 6x lower forces and operate over 10x wider energy bandwidth for personal safety and shipping applications (provisional pat. pending)

Independent Contractor, Engineering Design and Analysis FPrin, LLC — 2018–2024
Client-facing expert in numerical modeling for first-principles-based design and analysis

- Executed and communicated finite element analysis to international medical device client, resulting in retooling of 16 cavity production mold currently running at 10M units/month volume.
- Developed novel phase transformation model (whitepaper[2]) for shape memory alloys used in insulin pumps (sales >1M units/month); worked directly with Mathworks developers to implement model

Mechanical Engineer Lv.1, Lv.2 Triple Ring Technologies — 2014–2017
Numerical modeling, mechanical design, data collection and analysis, fixture design and validation, and client communication over 20 projects in 3 years.

- Led blank-slate design and fabrication effort of novel bolus delivery mechanism for miniature wearable insulin delivery device. US Patent 11007317
- Implemented feedback controller to regulate optical sensor temperature to $\pm 1°C$ in 510k approved oximetry device. International Patent WO2021142468A1
- Developed core competency in numerical simulation of multiphysics problems (heat transfer, fluid flow, continuum mechanics); delivered two 30 minutes technical presentations on rubber modeling and coupling global constraints to boundary integrals at the invitation of COMOSL Inc.

Grants & Awards

AB Nexus Spring 2021 Research Collaboration Grant Earned one year of funding by designing, fabricating, and mechanically characterizing multiphase composites alongside biological tissue

Matching Research Grant, Sandia National Laboratories Generated numerical and empirical results of high velocity impacts on tunable metamaterials, earning one year of research funding

Finalist, CU Boulder New Venture Challenge 2018 Analytical modeling and pitch delivery for electric drivetrain retrofitting system for freight industry.

COMPETENCES

Design Solidworks (expert certification), Inventor, Pro-E, Blender, Fusion360

Numerical Analysis Abaqus, COMSOL, Ansys, FEniCS, Simulink, Solidworks simulation

Fabrication Additive manufacturing, laser cutting, mill, lathe, aluminum casting

Data Processing & Presentation Matlab, git, MS Office Suite, LaTeX, Inkscape

SELECTED JOURNAL PUBLICATIONS

[1] Sophie Kirkman, Ingemar Schmidt, Lawrence T. Smith, Steven L. Zhang, Shane K. Mitchell, Soo Jin Adrian Koh, Philipp Rothemund, Christoph Keplinger "Maximizing the Energy Output of Soft Electrohydraulic Generators" *Advanced Science*, 2026

[2] Andrew K. Schulz, Lena V. Kaufmann, Lawrence T. Smith, Deepti S. Philip, Hilda David, Jelena Lazovic, Michael Brecht, Gunther Richter, and Katherine J. Kuchenbecker, "Functional gradients facilitate tactile sensing in elephant whiskers" *Science*, 2026.

[3] Lawrence Smith and Robert MacCurdy, "Digital Multiphase Composites via Additive Manufacturing" *Advanced Materials*, 2024.

[4] Lawrence Smith, Brandon Hayes, Kurtis Ford, Elizabeth Smith, Jesse Flores, Robert MacCurdy. "Tunable Metamaterials for Impact Mitigation" *Advanced Materials Technologies*, 2024.

[5] Lawrence Smith, Robert MacCurdy, "SoRoForge: End-to-End Soft Actuator Design." *IEEE Transactions on Automation Science and Engineering*, 2023.

[6] Travis Hainsworth, Lawrence Smith, Sebastian Alexander, and Robert MacCurdy. "A Fabrication Free, 3D Printed, Multi-Material, Self-Sensing Soft Actuator." *IEEE Robotics and Automation Letters*, 2020.

SELECTED CONFERENCE PROCEEDINGS

[7] Lawrence Smith, Travis Hainsworth, Zachary Jordan, Xavier Bell, and Robert MacCurdy. "A Seamless Workflow for Design and Fabrication of Pneumatic Soft Actuators." *IEEE CASE Proceedings, Lyon, France*, 2021. **Winner, Best Application Paper**

[8] Lawrence Smith, Travis Hainsworth, Jacob Haimes, and Robert MacCurdy. "Automated Synthesis of Bending Pneumatic Soft Actuators." *IEEE International Conference on Soft Robotics*, 2022.

[9] Lawrence Smith, Jacob Haimes, and Robert MacCurdy, "Stretching the Boundary: Shell Finite Elements for Pneumatic Soft Actuators." *IEEE International Conference on Soft Robotics*, 2022.

WHITEPAPERS

[10] Lawrence Smith. "A Novel Phase Transformation Model for Shape Memory Alloy Actuators" 2020.

MENTORSHIP

Research Advisor High-touch advisor to 15+ high school, undergraduate, and graduate engineering students. Generated three publications featuring these students as contributing authors.

INVITED TALKS

Invited Speaker Western Colorado University Spring 2023 *Computational Design of Soft Structures*

Invited Speaker Sandia National Laboratories RITS3 Lecture Series 2023 *Tunable Energy-Absorptive Metamaterials via Additive Manufacturing*

Workshop Lead International Conference on Intelligent Robots and Systems (IROS) 2022 *Accelerated Simulations of Soft Actuator Behavior*

Lecturer CU Boulder Summer Intensive Session 2020. *Introduction to Finite Elements*

Guest lecturer Rural Colorado high school science class Spring 2022. Taught air quality basics to freshman students and held symposium to exhibit students' projects.

Middle school robotics mentor FIRST VEX Challenge, Spring 2022, Spring 2020)

Co-Founder CU Boulder 3D Printing Club, Spring 2020

REFERENCES

Dr. Robert MacCurdy Ph.D. advisor, CU Boulder `maccurdy@colorado.edu`
Dr. Richard Regueiro Ph.D. committee, CU Boulder `richard.regueiro@colorado.edu`
Dr. Vani Sundaram Colleague, MPI-IS Stuttgart `vsun@is.mpg.de`
Peter Holst Colleague and CEO, FPrin LLC `pholst@fprin.com`